

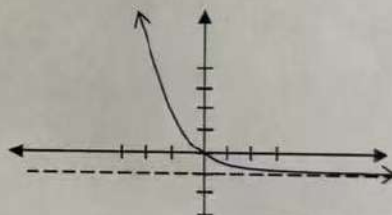
Instructions

- Non-graphing calculators may be used but not shared. Notebooks may not be used.
- Only methods taught in MHF4U1 at Bayview S.S. will be accepted. Show all work in the space provided and **show all necessary steps**. Leave answers **exact** unless otherwise specified. The use of cellphones, audio or video recording devices, digital music players or email or text-messaging devices during the assessment is prohibited.

KNOWLEDGE & UNDERSTANDING - [21 MARKS]

Multiple Choice: Write the **CAPITAL LETTER** corresponding to the correct answer on the line provided.
[1 Mark Each - 5 Marks Total]

- The y -intercept of $f(x) = \frac{2^{2(x+1)}}{4} + 1$ is D
A. $\frac{1}{4}$ B. 1 C. 0 D. 2
- An earthquake that is 100 times as intense as an earthquake measuring 6.8 on the Richter scale has A measure of B
A. 4.8 B. 8.8 C. 7.2 D. 6800
- Coke Zero has 4.17×10^{-8} moles of H^+ per litre. The pH level of Coke Zero is approximately D
A. 8 B. 4.17 C. 7.27 D. 7.38
- The range of $y = -\log_5(x) + 3$ is D
A. $y \in (3, \infty)$ B. $y \in (-\infty, 3)$ C. $y \in (-\infty, 0)$ D. $y \in (-\infty, \infty)$
- A possible equation of the graph shown is D
A. $y = 5^{x-1}$
B. $y = 5^x - 1$
C. $y = 5^{1-x}$
D. $y = 5^{-x} - 1$



- Sketch a properly labelled graph of $y = -\log_2(x-2)^2 + 3$. Clearly label all asymptote(s) and plot and label a minimum of 6 points on the graph. [4 Marks]

$x = \sqrt{2}^y$	
$\frac{1}{4}$	4
$\frac{1}{2}$	2
1	0
2	2
4	4
8	6

→

x	y
$\frac{1}{4}$	7
$\frac{1}{2}$	5
1	3
2	1
4	-1
8	-3

$$x-2=0$$

$$x=0+2$$

$$x=2$$

$$V.A. @ x=2$$

$$(x, y) \rightarrow (\frac{x}{k} + d, ay + c)$$

$$(x, y) \rightarrow (x+2, -y+3)$$

$$(\frac{1}{4}, -4) \rightarrow (\frac{9}{4}, 7)$$

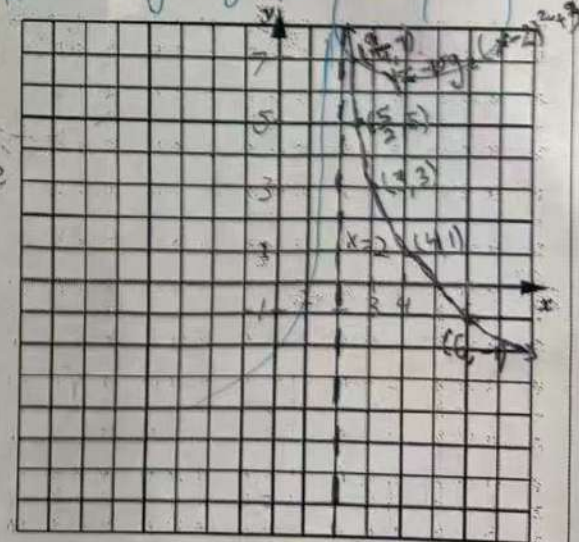
$$(\frac{1}{2}, -2) \rightarrow (\frac{5}{2}, 5)$$

$$(1, 0) \rightarrow (3, 3)$$

$$(2, 2) \rightarrow (4, 1)$$

$$(4, 4) \rightarrow (6, -1)$$

$$(8, 6) \rightarrow (10, -3)$$



7. Solve each of the following. Round your answers to 2 decimal places, if necessary. [6 Marks]

a. $5^{2x-9} = 18^{3x+4}$ [3]

$$5^{2x-9} = 18^{3x+4}$$

$$(2x-9)(\log 5) = (3x+4)(\log 18)$$

$$2x \log 5 - 9 \log 5 = 3x \log 18 + 4 \log 18$$

$$2x \log 5 - 3x \log 18 = 4 \log 18 + 9 \log 5$$

$$x(2 \log 5 - 3 \log 18) = 4 \log 18 + 9 \log 5$$

$$x = \frac{4 \log 18 + 9 \log 5}{2 \log 5 - 3 \log 18}$$

$$x \approx -4.78$$

b. $\log_9(x-5) = 1 - \log_9(x+3)$ [3]

$$\log_9(x-5) = 1 - \log_9(x+3)$$

$$\log_9(x-5) + \log_9(x+3) = 1$$

$$\log_9(x-5)(x+3) = 1$$

$$-1.5$$

8. Write $3 \log_{\sqrt{5}}(xy) + \log_{27}(x) - \frac{1}{3} \log_9(y^4)$ as a single fully simplified logarithm. [3 Marks]

$$-3$$

9. Determine inverse of $f(x) = 2 \log_7(3x-4) + 1$. [3 Marks]

$$f(x) = 2 \log_7(3x-4) + 1$$

$$\frac{x-1}{2} + 4 = 3y$$

$$y = 2 \log_7(3x-4) + 1$$

$$\frac{y-1}{2} + 4 = x$$

$$x = 2 \log_7(3y-4) + 1$$

$$y = \frac{2 \log_7(3x-4) + 1 - 1}{2} + 4$$

$$x-1 = 2 \log_7(3y-4)$$

$$\frac{x-1}{2} = \log_7(3y-4)$$

$$7^{\frac{x-1}{2}} = 3y-4$$

$$f^{-1}(x) = \frac{7^{\frac{x-1}{2}} + 4}{3}$$

$$-C$$

THINKING - [5 MARKS]

1. Solve $\log_x(x^2) + \log(\sqrt{1+x}) + 3 \log(\sqrt{1-x}) = \log(\sqrt{1-x^2})$. [5 Marks]

$$-5T$$

*** 2 Marks are awarded in the Communication Category for the use of correct mathematical form. ***

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APPLICATION - [18 MARKS]

1. Determine the domain, in interval notation, of $f(x) = \log_{2025}(x^3 - 5x^2 + 3x + 9)$. [4 Marks]

$$\begin{array}{r} f(x) = \log_{2025}(x^3 - 5x^2 + 3x + 9) \\ x^3 - 5x^2 + 3x + 9 > 0 \\ \begin{array}{r} x^3 - 5x^2 + 3x + 9 \\ x^3 - 6x^2 + 9x + 9 \\ \hline 9x + 9 \\ 9x + 9 \\ \hline 0 \end{array} \end{array}$$

$$x^2 - 6x + 9$$

$$= (x-3)^2$$

$$(x+1)(x-3)^2 > 0$$

$$x+1 > 0$$

$$x > -1$$

$$\cancel{x > 1}$$

Inadmissible

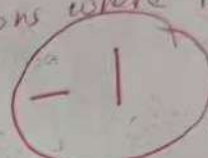
$$x-3 > 0$$

$$x > 0+3$$

$$x > 3$$

$$\text{Domain: } x \in (3, \infty)$$

solve inequality, check regions where inequality holds true



2. The decibel level of a baby's cry is 75 dB. The decibel level of the fire alarm at Bayview is 113 dB. How many times as intense is the Bayview fire alarm compared to the baby's cry? Round your answer to 1 decimal place. [3 Marks]

$$L_F - L_B = 10 \log\left(\frac{I_F}{I_B}\right)$$

$$113 - 75 = 10 \log\left(\frac{I_F}{I_B}\right)$$

$$38 \div 10 = \log\left(\frac{I_F}{I_B}\right)$$

$$3.8 = \log\left(\frac{I_F}{I_B}\right)$$

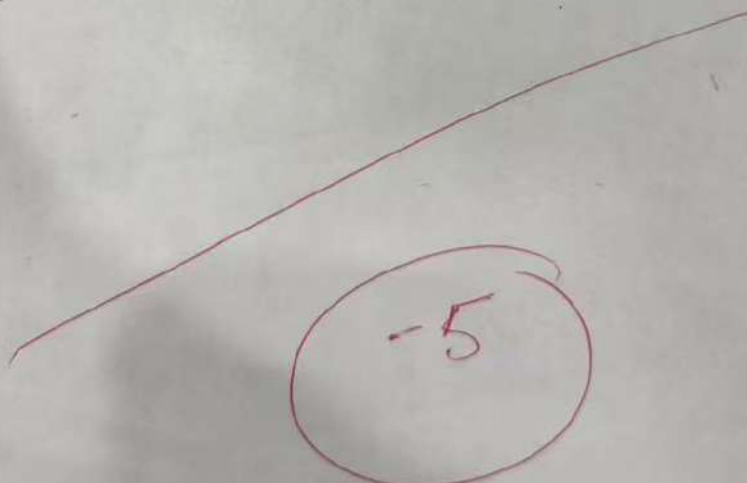
$$10^{3.8} = \frac{I_F}{I_B}$$

$$\frac{I_F}{I_B} \approx 6309.6$$

approximately

Bayview fire alarm is $\uparrow 6309.6$ times more intense than a baby's cry

3. Two pieces of art were purchased at the same time. The first piece of art costs \$995 and appreciates in value 5% every five years. The second piece of art costs \$795 and appreciates in value 8% every two years. How long will it take for the two pieces of art to be equal in value? Round your answer to 2 decimal places. [5 Marks]



4. The initial amount of Iodine-131 is 182 grams and it takes 23 days for 25 grams of it to remain. Determine the half-life of Iodine-131. Round your answer to 2 decimal places. [3 Marks]

Grishella please at least try to start question.

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5. Show that $\frac{1}{\log_7(x)} + \frac{1}{\log_6(x)} + \frac{1}{\log_2(x)} = \frac{1}{\log_{84}(x)}$. [3 Marks]

$$\frac{1}{\log_7(x)} + \frac{1}{\log_6(x)} + \frac{1}{\log_2(x)} = \frac{1}{\log_{84}(x)}$$

$$\log_x 7 + \log_x 6 + \log_x 2 = \log_x 84$$

$$\log_x (7 \times 6 \times 2) = \log_x 84$$

$$\log_x 84 = \log_x 84$$

$$\frac{1}{\log_{84}(x)} = \frac{1}{\log_{84}(x)}$$

THINKING - [5 MARKS]

1. Solve $6^{x^2+5} = \frac{1}{64^x \cdot 3^{6x}}$. [5 Marks]

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*** 2 Marks are awarded in the Communication Category for the use of correct mathematical form. ***